NCTU-EE IC Design LAB –Spring 2015

Final exam

6/17/2015 WED 13:30~15:15

Total: 115 points

PART 1 (42/115)

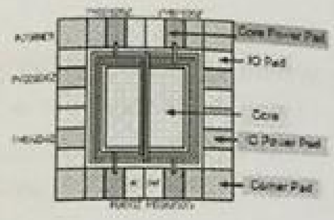
[APR]

(1) Please briefly describe what is inside LEF file? (3%)

(2) Please briefly describe what is inside LIB file? (3%)

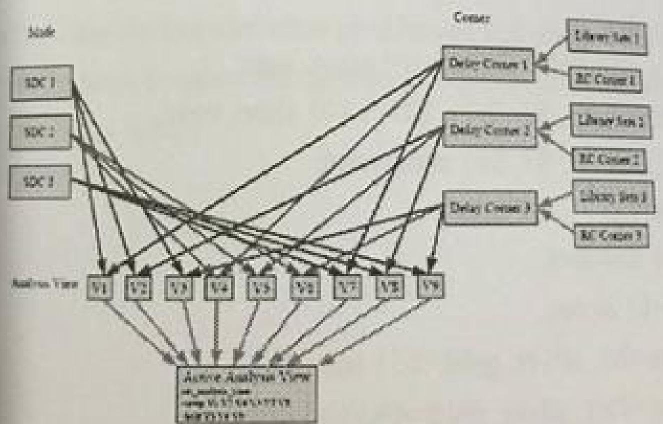
(3) What is the purpose of the "uniquify" command in the design flow? (3%)

(4) What's the job of corner PAD?(3%)



(5) What is the cause, and proper solution of IR drop? What will happen if the IR drop is severe?(3%)

(6) Why MMMC ? (3%)



(7) When we are adding IO filler, is the following order of text command correct? Why or why not?(3%)

encounter #> addIoFiller -cell PFILL_01 -prefix IOFILLER

encounter #> addIoFiller -cell PFILL_1 -prefix IOFILLER

encounter #> addIoFiller -cell PFILL_9 -prefix IOFILLER

encounter #> addIoFiller -cell PFILL -prefix IOFILLER –fillAnyGap

(8) When there are dangling wires(LVS problem),what is the hot key to trim it?(3%)

(9) Please give some examples of the difference between SOC encounter and IC Compiler.(6%)

[DFT&scan chain]

(10) In fault model, we usually use single fault assumption, please explain why?(4%)

(11) Why DFT & ATPG needed in Gate (logic)level?(4%)

(12) What are the disadvantages of full scan compared to partial scan?(4%)

PART 2 (24/115)

(13) List at least two enhancement in design using system verilog , and explain what is the advantage. (4%)

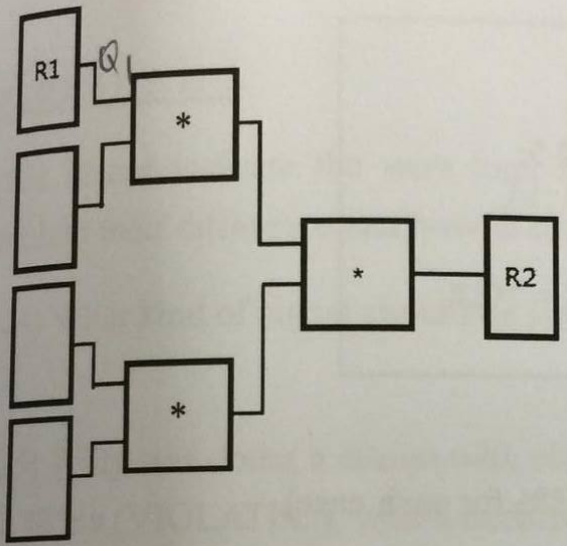
(14) List at least two enhancement in verification using system verilog, and explain what is the advantage. (4%)

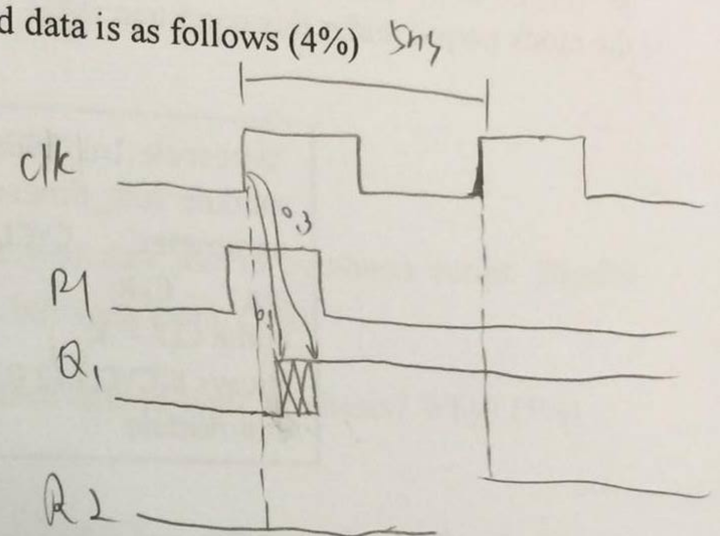
(15) Please check if there are syntax errors in the following code. Correct the error if there are existed errors.(8%)

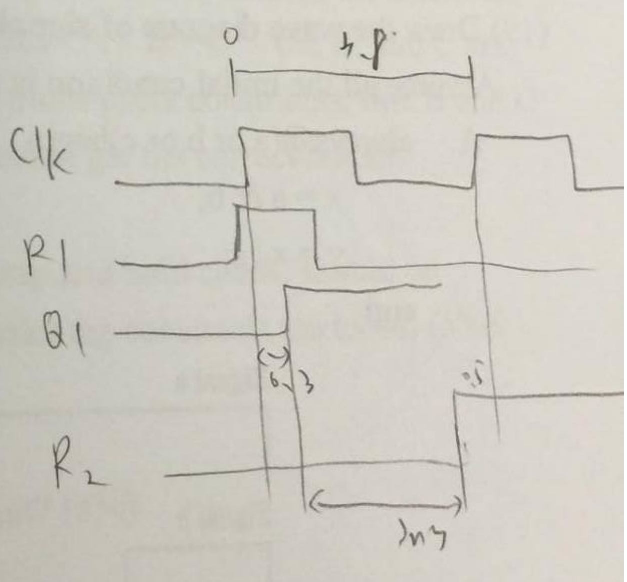
1. **enum** {IDLE, XX='x, S1=2'b01, S2=2'b10} state, next;
2. **enum integer** {IDLE, XX='x, S1='b01, S2='b10} state, next;
3. **enum integer** {IDLE, XX='x, S1, S2} state, next;
4. **enum** {a=3, b=7, c} alpha;
5. **enum bit** [0:0] {a,b,c} alphabet;
6. **enum** {a=0, b=7, c, d=8} alpha;
7. **enum bit** [3:0] {bronze='h3, silver, gold='h5} medal4;
8. **enum bit** [3:0] {bronze=4'h3, silver, gold=4'h5} medal4;

(16) Please briefly explain "setup time" and "hold time".(4%)

(17) There is a simple design and the related data is as follows (4%)







CLK period = 5ns

Setup time for each reg = 0.5ns

Hold time for each reg = 0 ns

t_{pcq} = 0.3ns

t_{ccq} = 0.1ns

Multiplier delay = 3ns

CLK skew = 0.6ns

Is there any timing violation problem in this design? If your answer is yes, please describe how to solve this problem. Besides, show the range of combinational logic delay (t_d) as well.

PART 3 (24/115)

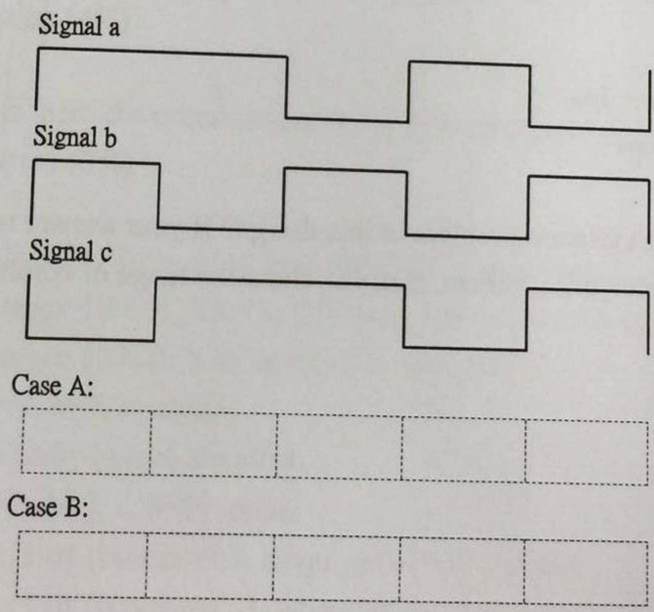
(18) Some mistake occurs when generating the clock, please describe the problem and the clock period under this condition.(6%)

```
`timescale 1ns/100ps
module Test_timescale;
parameter     CYCLE = 2.5;
reg       CLK;          1.25
initial CLK = 1;
always #(CYCLE/2.0) CLK = ~CLK;
endmodule
```

(19) Draw the wave diagram of **signal y** below. (3% for each case)
Assume all the initial condition is zero.

A.
```
always@(a or b or c)begin
    x = a & b;
    y = x | c;
end
```

B.
```
always@(a or b or c)begin
    x <= a & b;
    y <= x | c;
end
```



(20) What are the differences between blocking and non-blocking assignment description? (2%)

(21) Please briefly describe the "dynamic clock gating" and "data gating method" and draw the corresponding logic diagram. (4%)

(22) Give one design method for reducing dynamic or static power in architecture/RTL-level, gate-level and transistor-level, respectively (3 methods in total). Also briefly describe these methods. (6%)

PART 4 (25/115)

(23) Please indicate the wire load mode you can use in synthesis script. Briefly explain their difference and how to choose between them. (5%)

(24) What kind of signal should be set as ideal nets in logic synthesis? Why? (5%)

(25) Willy was doing a design with clock period = 5ns. After synthesis, the slack was -0.725ns (VIOLATED), which occurred on the path " A = B ÷ C ". (A, B, and C are the names of registers). So he decided to add the multi-cycle constraints, that B and C would be stable for at least two cycles so that A could get the correct output.

Please list the multi cycle constraints (include setup and hold check, should be complete) to fix the slack of this path. (Do not make the constraint too loose) (5%)

(26)

a. How do you design Multi-Clock Domain circuit? (5%)

b. Please describe your design strategy during logic synthesis and gate level simulation. (5%)